

DEQEN, China

WMO: 564440

Lat: 28.4886N

Lon: 98.9069E

Elev: 10892

StdP: 9.76

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	19.5	-1.4	7.7	34.6	1.8	9.1	33.3	14.8	40.5	12.8	39.3	5.3	270	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	70.4	54.5	68.2	54.1	66.3	53.6	56.8	65.5	56.2	64.4	55.5	63.1	5.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.3	95.1	58.4	53.6	92.8	58.0	53.0	90.5	57.7	29.8	66.0	29.3	65.0	28.7	63.4	67.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	9.7	8.4	13.0	75.7	2.8	2.7	10.9	77.7	9.3	79.2	7.7	80.7	5.7	82.7		
(5)			10.1	58.8	2.3	1.9	8.4	60.2	7.0	61.3	5.7	62.4	4.0	63.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	30.7	33.4	37.5	43.1	50.4	56.9	58.0	57.7	55.0	47.1	38.8	32.7	(6)	
	DBStd	10.67	4.06	4.17	3.91	4.19	4.36	3.29	3.23	2.77	3.65	4.22	3.52	3.53	(7)	
	HDD50	2703	598	464	389	209	49	1	1	0	8	110	337	537	(8)	
	HDD65	7239	1063	884	854	656	453	242	217	228	299	554	787	1002	(9)	
	CDD50	939	0	0	0	2	61	209	249	238	159	21	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	8	0	0	0	0	1	3	2	2	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	3.9	4.5	4.5	4.2	4.3	4.2	3.5	3.0	3.0	3.3	4.1	4.4	4.5	(14)	
	PrecAvg	28.5	0.6	1.0	2.6	2.8	2.6	3.0	5.6	4.7	2.8	1.9	0.7	0.2	(15)	
(16) Precipitation	PrecMax	40.1	2.2	2.9	7.5	9.2	6.4	5.5	9.6	7.0	6.5	5.7	4.1	1.0	(16)	
	PrecMin	20.8	0.0	0.2	0.6	0.3	0.4	1.6	2.4	2.4	0.5	0.3	0.0	0.0	(17)	
	PrecStd	5.0	0.5	0.8	1.7	1.9	1.6	1.0	1.5	1.4	1.4	1.2	0.9	0.2	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.0	53.9	56.8	62.7	69.7	73.5	72.5	72.5	70.2	65.2	55.3	49.9	(19)	
		MCWB	33.3	35.8	39.1	44.4	49.5	54.9	55.9	55.4	54.0	49.4	38.8	33.3	(20)	
	2%	DB	46.8	49.8	53.0	59.2	66.2	70.0	69.7	69.0	66.7	61.5	52.9	46.7	(21)	
		MCWB	31.7	34.1	37.6	42.9	48.9	53.7	55.7	55.2	53.1	47.6	37.9	32.2	(22)	
	5%	DB	43.3	46.4	50.1	56.2	63.6	67.5	67.2	66.5	64.5	58.7	50.7	44.2	(23)	
		MCWB	29.7	32.9	36.4	41.8	48.1	53.3	55.6	55.1	52.6	46.1	37.2	31.0	(24)	
	10%	DB	40.2	43.4	47.0	53.2	60.8	65.2	64.9	64.2	62.2	56.1	48.2	41.8	(25)	
		MCWB	28.6	31.6	35.3	40.8	47.2	53.1	55.1	54.7	52.2	44.6	36.0	30.0	(26)	
	0.4%	WB	35.3	37.6	41.6	46.4	52.8	57.1	58.2	57.6	56.1	51.4	42.8	35.6	(27)	
		MCDB	47.4	48.2	51.2	59.3	63.6	66.7	67.9	66.5	65.1	60.3	48.7	46.4	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.5	35.4	39.4	44.6	51.2	56.0	57.1	56.6	54.8	49.9	40.8	33.4	(29)	
		MCDB	44.1	46.0	48.3	55.8	61.1	65.4	66.0	65.0	63.1	56.8	47.5	43.5	(30)	
	5%	WB	30.7	33.8	37.9	43.2	49.8	55.2	56.4	55.8	53.9	48.6	39.0	32.0	(31)	
		MCDB	40.9	44.0	46.5	52.7	59.5	64.1	64.3	63.6	61.6	54.8	47.5	41.9	(32)	
	10%	WB	29.2	32.3	36.5	41.8	48.5	54.3	55.7	55.1	53.0	47.4	37.4	30.7	(33)	
		MCDB	38.6	41.6	44.5	50.4	57.3	62.3	62.7	62.1	59.8	53.0	46.3	40.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.7	18.4	16.7	16.9	17.2	15.5	13.5	14.0	14.0	17.4	18.8	19.5	(35)	
		MCDBR	23.4	25.0	23.3	23.1	23.5	20.4	17.8	18.2	17.9	20.6	22.4	22.1	(36)	
		MCWBR	12.3	12.7	10.9	9.7	8.7	7.0	6.3	6.6	6.5	8.5	10.6	11.5	(37)	
	5% WB	MCDBR	20.8	21.5	18.7	19.4	18.2	15.5	14.4	14.8	14.5	15.4	18.1	19.9	(38)	
		MCWBR	11.5	11.4	9.6	9.1	7.3	6.2	5.7	5.9	5.7	6.6	9.4	10.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.242	0.279	0.348	0.346	0.312	0.300	0.307	0.316	0.292	0.259	0.234	0.222	(40)	
	taud		2.555	2.445	2.234	2.321	2.517	2.650	2.634	2.572	2.631	2.711	2.707	2.678	(41)	
	Ebn at Noon		324	317	298	301	309	311	308	306	313	322	327	330	(42)	
	Edh at Noon		27	33	43	41	34	30	30	32	29	25	23	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		1273	1409	1483	1526	1663	1639	1505	1485	1393	1385	1324	1264	(44)	
	RadStd		138	118	180	167	128	127	129	139	90	113	107	79	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.49	N/A	N/A	+0.90	N/A	N/A	-160	-311	+131	N/A	(46)
(47)	Regional (2 neighbors)	+0.79	N/A	N/A	+1.39	N/A	N/A	-133	-292	+187	N/A	(47)

Nomenclature: See separate page